Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Models the motion of the ball by forming an equation of motion	AO3.1b	M1	$m\dfrac{d^2x}{dt^2} = (-12.5mx)\times 2$
	Uses SHM equations to form model for displacement	AO3.1a	M1	$\Rightarrow \dfrac{d^2x}{dt^2} = -25x$ $\therefore x = A\sin(5t)$
	Uses initial condition to find the constant	AO1.1a	M1	$\Rightarrow \dot{x} = 5A\cos(5t)$
	Obtains correct value for constant	AO1.1b	A1	
	Interprets 'their' value to find minimum distance from P	AO3.2a	A1F	when $t = 0$, $\dot{x} = 0.75$ so $0.75 = 5A$ $A = 0.15$ Hence $x = 0.15\sin(5t)$ Max displacement = 0.15 metres from O, when $\sin(5t) = \pm 1$, so minimum distance from P is 0.75 metres
(b)	Identifies a correct limitation of the model for example friction between ball and the surface or damping effect due to air	AO3.5b	B1	It is unlikely that the surface is perfectly smooth so friction will be acting. The ball will be likely to travel a smaller distance before coming to rest and the minimum distance of the ball from P may actually be greater than that calculated in part **(a)**.
	Correctly infers whether the distance is too big or too small based on the limitation they have identified. Accept any well-reasoned inference.	AO2.2b	R1	
				Total 7 marks

Q2.

(a) $m\ddot{x} = -4mn^2x - 2mk\dot{x}$

M1A1

$\ddot{x} + 2k\dot{x} + 4n^2x = 0$

AG

A1

3

(b) (i) $k = n$

$p^2 + 2np + 4n^2 = 0$

$(p + n)^2 + 3n^2 = 0$

M1

$p = -n \pm n\sqrt{3}\mathrm{i}$

A1

$x = \mathrm{e}^{-nt}(A\cos\sqrt{3}nt + B\sin\sqrt{3}3nt)$

ft provided in correct form

A1F

$t = 0, x = a \Rightarrow a = A$

A1

$\dot{x} = \mathrm{e}^{-nt}(-\sqrt{3}nA\sin\sqrt{3}nt + \sqrt{3}nB\cos\sqrt{3}nt)$

$-n\mathrm{e}^{-nt}(A\cos\sqrt{3}nt + B\sin\sqrt{3}3nt)$

m1

$t = 0, \dot{x} = 0 \Rightarrow 0 = \sqrt{3}nB - nA$

$B = \frac{a}{\sqrt{3}}\left(=\frac{\sqrt{3}a}{3}\right)$

A1F

$x = \mathrm{e}^{-nt}\left(a\cos\sqrt{3}nt + \frac{\sqrt{3}a}{3}\sin\sqrt{3}nt\right)$

AG

A1F

7

(ii) $x = 0,\ a\cos\sqrt{3}nt + \frac{\sqrt{3}}{3}a\sin\sqrt{3}nt = 0$

Condone verification with $t > 0$

M1

$(e^{-nt} \neq 0)$

$\tan\sqrt{3}nt = -\sqrt{3}$
AG

M1A1

3

(c) (i) $k = 2n$

$p^2 + 4np + 4n^2 = 0$

M1

$(p + 2n)^2 = 0$

M1

$x = e^{-2nt}(A + Bt)$

A1

3

(ii) critical damping

A1F

1

[17]

Q3.

(a) $x = A\cos 4t + B\sin 4t$
$\dot{x} = -4A\sin 4t + 4B\cos 4t$

B1

$\ddot{x} = -16A\cos 4t - 16B\sin 4t$

B1

Alt: *$m^2 + 16 = 0$* *B1*
$m = \pm 4i$ *B1*
$x = A\cos 4t + B\sin 4t$ *M1A1*

Substitute into $\ddot{x} + 16x = 0$

M1

Satisfactory conclusion

A1

4

(b) $t = 0$, $\dot{x} = 0$: $0 = 0 + 4B \rightarrow B = 0$
AG

B1

1

(c) $x = A\cos 4t$

$t = \dfrac{\pi}{12}, x = h: \; h = A\cos\dfrac{\pi}{3}$

M1

$A = 2h$

A1

2

(d) $\dot{x} = -8h\sin 4t$

Max speed = $8h$ m s^{-1}

B1

1

(e) $F = m(-32h\cos 4t)$

M1

$|F_{max}| = 32hm$ N

A1

2

[10]

Q4.

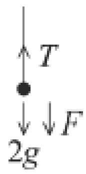
(a)

$2g = \dfrac{49 \times e}{0.5}$ M1

$e = 0.2$ metres A1

2

(b)



$2\ddot{x} = 2g + F - T$ M1A1

$2\ddot{x} = 2g + 12\cos nt - \dfrac{49(x+0.2)}{0.5}$ *Tension ft (a)* B1F

$2\ddot{x} = 2g + 12\cos nt - 98x - 19.6$

ft (a)

A1F

$\ddot{x} + 49x = 6\cos nt$

AG

A1

5

(c) $n = 5$ PI, $x = A\cos 5t + B\sin 5t$

Accept cos term only

M1

$\dot{x} = -5A\sin 5t + 5B\cos 5t$

A1

$\ddot{x} = -25A\cos 5t - 25B\sin 5t$

A1

Subs:

$-25A\cos 5t - 25B\sin 5t + 49A\cos 5t + 49B\sin 5t = 6\cos 5t$

$B = 0,\quad A = \frac{1}{4}$

A1

A. eqn: $m^2 + 49 = 0$ $m = \pm 7\mathrm{i}$

M1

C.F.: $x = C\cos 7t + D\sin 7t$

A1

Gen sol: $x = C\cos 7t + D\sin 7t + \frac{1}{4}\cos 5t$

M1

$t = 0, x = 0:\quad 0 = C + \frac{1}{4}\quad C = -\frac{1}{4}$

A1

$\dot{x} = -7C\sin 7t + 7D\cos 7t - \frac{5}{4}\sin 5t$

m1

$t = 0, \dot{x} = 0:\quad D = 0$

$x = \frac{1}{4}(\cos 5t - \cos 7t)$

A1

10

(d) Resonance occurs when $n = 7$

B1

1

[18]

Q5.

(a) Natural length of AP is $4a$ and natural length of BP is $2a$

When particle is x from equilibrium position:

Tension in AP is $\dfrac{4mn^2a(2a+x)}{4a}$

M1A1

Tension in BP is $\dfrac{4mn^2a(a-x)}{2a}$

M1A1

In general position, using $F = ma$:

$$m\frac{d^2x}{dt^2} = \frac{4mn^2(a-x)}{2a} - \frac{4mn^2a(2a+x)}{4a} - 2mn\frac{dx}{dt}$$

M1A1

$$m\ddot{x} = 2mn^2a - 2mn^2x - 2mn^2a - mn^2x - 2mn\dot{x}$$

$$\frac{d^2x}{dt^2} + 2n\frac{dx}{dt} + 3n^2x = 0$$

AG

A1

7

(b) [Substituting $x = Ae^{pt}$]

$p^2 + 2p + 3 = 0$

M1A1

$p = -1 \pm \sqrt{2}\,i$

A1

General solution is:

$x = e^{-t}(A\cos\sqrt{2}t + B\sin\sqrt{2}t)$

A1

When $t = 0$, $x = \frac{1}{2}a \Rightarrow \frac{1}{2}a = A$

B1

Differentiating:

$$\frac{dx}{dt} = -e^{-t}(A\cos\sqrt{2}t + B\sin\sqrt{2}t) + e^{-t}(-A\sqrt{2}\sin\sqrt{2}t + B\sqrt{2}\cos\sqrt{2}t)$$

M1A1

When $t = 0$, $\frac{dx}{dt} = 0$

$\Rightarrow 0 = -A + \sqrt{2}B$

$A = \frac{1}{2}a,\ B = \frac{1}{2\sqrt{2}}a$

A1

$$x = ae^{-t}(\frac{1}{2}\cos\sqrt{2}t + \frac{1}{2\sqrt{2}}\sin\sqrt{2}t)$$

8

[15]

Q6.

(a) Using $F = ma$,

$2m\ddot{x} = 2mg - T - 4mnv$

3 terms at least correct

M1

$= 2mg - 8mn^2a \,.\, \frac{x}{a} - 4mnv$

A1

$m\ddot{x} + 4mn^2x + 2mn\dot{x} = mg$

i.e. $\frac{d^2x}{dt^2} + 2n\frac{dx}{dt} + 4n^2x = g$

A1

3

(b) Substituting $x = Ae^{pt}$

$p^2 + 2np + 4n^2 = 0$

M1

$p = (-2n \pm \sqrt{4n^2 - 16n^2}) \div 2$

$= -n \pm \sqrt{3}in$

A1

C.F. is

$x = e^{-nt}(A\cos\sqrt{3}nt + B\sin\sqrt{3}nt)$

Dep M1

B1ft

P.I. is x = $\frac{g}{4n^2}$

B1

General solution is x =

$e^{-nt}(A\cos\sqrt{3}nt + B\sin\sqrt{3}nt) + \frac{g}{4n^2}$

ft dep M1first line (b)

M1

At least one of CF, PI correct

A1ft

When $t = 0$, $x = 0$, $\Rightarrow$ $0 = A + \frac{g}{4n^2}$

M1

$\therefore\ A = -\frac{g}{4n^2}$

A1

When $t = 0$, $\frac{dx}{dt} = 0$;

$\frac{dx}{dt} = -ne^{-nt}(A\cos\sqrt{3}t + B\sin\sqrt{3}t) +$

M1

$e^{-nt}(-\sqrt{3}nA\sin\sqrt{3}nt + \sqrt{3}nB\cos\sqrt{3}nt)$

A1

$$\Rightarrow 0 = -nA + \sqrt{3}\,nB$$

$$\therefore B = \frac{1}{\sqrt{3}}A = -\frac{g}{4\sqrt{3}n^2}$$

A1

$$\therefore x = \frac{g}{4n^2} - \left(\frac{g}{4n^2}\cos\sqrt{3}\,nt + \frac{g}{4\sqrt{3}n^2}\sin\sqrt{3}\,nt\right)e^{-nt}$$

A1

2

(c) $$\frac{dx}{dt} = ne^{-nt}\left(\frac{g}{4n^2}\cos\sqrt{3}nt + \frac{g}{4\sqrt{3}n^2}\sin\sqrt{3}nt\right)$$

M1

$$-e^{-nt}\left(-\frac{\sqrt{3}g}{4n^2}\sin\sqrt{3}nt + \frac{g}{4n}\cos\sqrt{3}nt\right)$$

A1

$$\frac{dx}{dt} = 0 \text{ when}$$

$$\cos\sqrt{3}nt + \frac{1}{\sqrt{3}}\sin\sqrt{3}nt + \sqrt{3}\sin\sqrt{3}nt$$

$$-\cos\sqrt{3}nt = 0$$

M1

$$\sin\sqrt{3}nt = 0$$

First time at rest is when $t = \dfrac{\pi}{\sqrt{3}n}$

A1

4

[19]

Q7.

(a) $T = 10(x - 0.5)$ or $-10(x - 0.5)$

M1A1

2

(b) $10(x - 0.5) - 2v = 1\dfrac{dv}{dt}$

Newton's 2nd law

M1A1F

$$\frac{dv}{dt} + 2v - 10x = -5$$

AG

A1

3

(c) (i) The rate of change of the length of the spring = speed of A relative to B

OE

E1

1

$\therefore\ \dot{x} = 1 - v$

(ii) $-\ddot{x} + 2(1 - \dot{x}) - 10x = -5$

M1

$\ddot{x} + 2\dot{x} + 10x = 7$

AG

A1

2

(d) Auxiliary equation: $m^2 + 2m + 10 = 0$

M1

$m = -1 \pm 3i$

A1

CF $x = e^{-t}(A \sin 3t + B \cos 3t)$

M1

PI $x = \alpha$

$\dot{x} = \ddot{x} = 0$

M1

A1F

$\alpha = \frac{7}{10}$

GS $x = e^{-t}(A \sin 3t + B \cos 3t) + \frac{7}{10}$

A1F

6

(e) $t \to \infty$

$e^{-t} \to 0$

$x \to \frac{7}{10}$

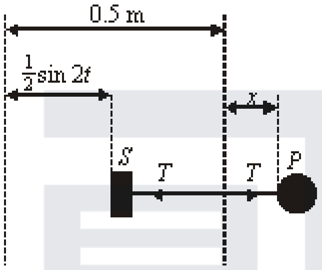
M1A1F

2

[16]

Q8.

(a)



B1

1

(b) $T = 4.5(\frac{1}{2}\sin 2t - x)$

M1

$T = 0.5\ddot{x}$

A1

$0.5\ddot{x} = 4.5(\frac{1}{2}\sin 2t - x)$

Newton's 2nd law

M1

$\ddot{x} + 9x = \frac{9}{2}\sin 2t$

AG

A1

4

(c) Auxiliary eq^n. $m^2 + 9 = 0$

M1

$m = \pm 3i$

A1

C.F. $x = A\text{Cos}3t + B\text{Sin}3t$

B1

For P.I. $x = p\cos 2t + q\sin 2t$

$\dot{x} = -2p\sin t + 2q\cos 2t$

$\ddot{x} = -4p\cos 2t - 4q\sin 2t$

M1

$5p\cos 2t + 5q\sin 2t = \frac{9}{2}\sin 2t$

M1

$p = 0, q = \frac{9}{10}$

For both p and q

A1F

G.S. is

$x = A\cos 3t + B\sin 3t + \frac{9}{10}\sin 2t$

A1F

$x = \dot{x} = 0, t = 0$

$\therefore A = 0$

A1F

$\dot{x} = 3B\cos 3t + \frac{9}{5}\cos 2t$

$\therefore B = -\frac{3}{5}$

A1F

$x = -\frac{3}{5}\sin 3t + \frac{9}{10}\sin 2t$

9

(d) $\dot{x} = -\frac{9}{5}\cos 3t + \frac{9}{5}\cos 2t$

M1

$= -\frac{9}{5}\cos\frac{6\pi}{5} + \frac{9}{5}\cos\frac{4\pi}{5}$

$= 0$

A1

2

[16]

Q9.

(a) Auxiliary equation: $m^2 + 2m + 5 = 0$

M1

$m = -1 \pm 2i$

A1

CF $\theta = e^{-t}(A\cos 2t + B\sin 2t)$

A1F

For PI $\theta = \alpha$

correct usage

M1

$\dot{\theta} = 0$

$5\alpha = 0.3 \Rightarrow \alpha = 0.06$

A1

GS = CF + PI

B1F

6

(b) $\dot{\theta} = -e^{-t}(A\cos 2t + B\sin 2t) + e^{-t}(-2A\sin 2t + B\cos 2t)$

Differentiating their GS (must be correct form)

M1 A1F

$0.2 = e^0(A\cos 0 + B\sin 0) + 0.06$

$A = 0.14$

ft from part (a)

A1F

$0 = -e^0(A\cos 0 + B\sin 0) + e^0(-2A\sin 0 + B\cos 0)$

$0 = -A + 2B \quad 0 = -0.14 + 2B$

$B = 0.07$

$\theta = e^{-t}(0.14\cos 2t + 0.007\sin 2t) + 0.06$

A1F A1F

5

(c) As $t \to \infty$, $e^{-t} \to 0$ and

$e^{-t}(0.14\cos 2t + 0.07\sin 2t) \to 0$

Correct from with A or B ≠ 0

M1

$\therefore\ \theta \to 0.06$

A1F

2

[13]

Q10.

(a) $2 \times 9.8 = \frac{\lambda}{0.5} \times 0.2$

Equilibrium considered to form equation in λ

M1

$\lambda = \frac{9.8}{0.2} = 49$ N

Correct equation

A1

Correct λ

A1

3

(b) $T = \frac{49x}{0.5} + 2g$

Equation for tension with two terms

M1

Correct equation

A1

$2\frac{d^2x}{dt^2} = 2g - (98x + 2g)$

Use of $F = m\frac{d^2x}{dt^2}$

M1

$\frac{d^2x}{dt^2} = -\frac{98}{2}x = -49x$

Correct equation

A1

ag *Correct result from correct working*

A1

5

(c) Period = $\frac{2\pi}{\sqrt{49}}$

Finding period

M1

Correct period

A1

$t = \frac{1}{4} \times \frac{2\pi}{7} = \frac{\pi}{14}$ = 0.224 seconds

Dividing period by 4

M1

Correct time

A1

4

[12]

Q11.

(a) Substituting $x = A\mathrm{e}^{nt}$ into CF

$n^2 + 6n + 10 = 0$

$n = \frac{-6 \pm \sqrt{36-40}}{2}$

$-3 \pm \mathrm{i}$

M1

$\therefore$ CF is $x = \mathrm{e}^{-3t}(A\cos t + B\sin t)$

M1A1

PI: $x = C\sin 3t + D\cos 3t$

M1 only for PI section if only Csin3t or Dcos3t used

M1

$\frac{\mathrm{d}x}{\mathrm{d}t} = 3C\cos 3t - 3D\sin 3t$

$\frac{d^2x}{dt^2}$ = –9Csin3t – 9 D cos3t

Substituting into

$\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 10x = 325 \sin 3t$

$-9C\sin 3t - 9D\cos 3t + 18C\cos 3t$
$-18D\sin 3t + 10C\sin 3t + 10D\cos 3t$

M1A1

$= 325\sin 3t$

(sin t) $C - 18D = 325$
(cos t) $D + 18C = 0$
$325C = 325$
$C = 1$ B1
$D = -18$

B1

$\therefore$ PI is $x = \sin 3t - 18\cos 3t$
$x = e^{-3t}(A\cos t + B\sin t) + (\sin 3t - 18\cos 3t)$
When $x = 0$, $t = 0$
$\Rightarrow 0 = A - 18$
$\Rightarrow A = 18$

B1

$\frac{dx}{dt} = -3e^{-3t}(A\cos t + B\sin t) +$

$e^{-3t}(B\cos t - A\sin 3t) + 3\cos 3t + 54\sin 3t$

When $t = 0$, $\frac{dx}{dt} = 0$

$\Rightarrow 0 = -3A + B + 3$
$B = 51$

B1

$\therefore$ $x = (18\cos t + 51\sin t)e^{-3t} +$
$\sin 3t - 18\cos 3t$

Accept $x = 54e^{-3t}\cos(t - 1.232)$
$+ (\sin 3t - 18\cos 3t)$
allow 54, 54.1, 54.08

B1

11

(b) When t is large, $x \approx \sin 3t - 18\cos 3t$

B1

This is periodic with period $\frac{2\pi}{3}$

B1

and amplitude $5\sqrt{13}$

Accept $\sqrt{325}$

B1

3

[14]

Q12.

(a) $mg = \frac{\lambda}{l} \times \frac{l}{4}$

Consideration of forces in equilibrium

M1

$\lambda = 4mg$

ag *Correct λ from correct working*

A1

2

(b) (i) $m\frac{d^2x}{dt^2} = mg - T$

$= mg - \frac{4mg}{l}\left(x + \frac{l}{4}\right)$

Two term expression for T

M1

Correct expression for T

A1

Three term equation of motion

M1

$= -\frac{4mg}{l}x$

Simplifying

m1

$$\frac{d^2x}{dt^2} = -\frac{4g}{l}x$$

***ag** Correct result from correct working*

A1

5

(ii) Period $= 2\pi\sqrt{\frac{1}{4g}} = \pi\sqrt{\frac{1}{g}}$

Identifying ω

M1

Correct period

A1

2

[9]

Q13.

(a) When displacement is x,

Tension $= \frac{\lambda x}{l} = \frac{4mn^2 a.x}{2a}$

$= 2mn^2x$

B1

Using $F = ma$,

$$m = \frac{d^2x}{dt^2} = mg - 2mn^2x - 2mn\frac{dx}{dt}$$

M1 A1

$$\frac{d^2x}{dt^2} + 2n\frac{dx}{dt} + 2n^2x = g$$

A1

4

(b) CF $x = Ae^{pt}$

$p^2 + 2np + 2n^2 = 0$

M1

$$p = \frac{-2n \pm \sqrt{4n^2 - 8n^2}}{2}$$

$= -n \pm ni$

A1

$\therefore\ x = e^{-nt}(A\cos nt + B\sin nt)$

A1

$$\text{PI } x = \frac{g}{2n^2}$$

B1

$$x = e^{-nt}(A\cos nt + B\sin nt) + \frac{g}{2n^2}$$

M1

When $t = 0$, $x = 0$,

$$A = -\frac{g}{2n^2}$$

B1

$$\frac{dx}{dt} = -ne^{-nt}(A\cos nt + B\sin nt) +$$

M1

$$e^{-nt}(-nA\sin nt + nB\cos nt)$$

When $t = 0,\ \dfrac{dx}{dt} = 0$

$\Rightarrow -nA + nB = 0$

$$B = -\frac{g}{2n^2}$$

A1

$$x = \frac{g}{2n^2}\{1 - e^{-nt}(\cos nt + \sin nt)\}$$

A1

9

[13]

Q14.

(a) $T_{AP} = \dfrac{\lambda.3a}{4a} = \dfrac{3}{4}\lambda$

$T_{PB} = 4mg.\dfrac{a}{2a} = 2\text{mg}$

Either

B1

Using $F = ma$ vertically

M1

$mg + T_{PB} = T_{AP}$

$\therefore\ mg + 2mg = \frac{3}{4}\lambda$

A1

$\lambda = 4mg$

A1

4

(b) (i) When particle is moved a distance x below the equilibrium position, forces acting on it are

$mg,\ T_{AP} = \frac{\lambda.(3a + x)}{4a} = \frac{mg(3a + x)}{a}$,

M1

$T_{PB} = 4mg.\frac{(a - x)}{2a} = \frac{2mg}{a}(a - x)$

and resistance $\frac{1}{3}mk\dot{x}$

All four forces

M1

[forces 2 and 4 are upwards]

Using $F = ma$ vertically downwards

$m\ddot{x} = mg + T_{PB} - T_{AP} - \frac{1}{3}mk\dot{x}$

Dependent on both M1 above

m1

$m\ddot{x} = mg + \frac{2mg}{a}(a - x) - \frac{mg(3a + x)}{a} - \frac{1}{3}mk\dot{x}$

A1

$\ddot{x} - g - \frac{2g}{a}(a - x) + \frac{g(3a + x)}{a} + \frac{1}{3}k\dot{x} = 0$

$$\ddot{x} + \tfrac{1}{5}k\dot{x} + \frac{3gx}{a} = 0$$

A1

$$10\frac{d^2x}{dt^2} + 2k\frac{dx}{dt} + 5k^2x = 0$$

A1

6

(ii) Substituting $x = Ae^{nt}$,

$10n^2 + 2kn + 5k^2 = 0$

$$n = \frac{-2k \pm \sqrt{4k^2 - 200k^2}}{20}$$

M1

$= \frac{1}{10}(-k \pm 7ki)$

A1

$$x = e^{-\frac{k}{10}t}(A\cos \tfrac{7}{10}kt + B\sin \tfrac{7}{10}kt)$$

M1 A1ft

When $t = 0$, $x = \frac{a}{2}$, $A = \frac{a}{2}$

B1

Differentiating

$$\frac{dx}{dt} = -\frac{k}{10}e^{-\frac{k}{10}t}(A\cos \tfrac{7}{10}kt + B\sin \tfrac{7}{10}kt) +$$

$$e^{-\frac{k}{10}t}(-\tfrac{7}{10}kA\sin \tfrac{7}{10}kt + \tfrac{7}{10}kB\cos \tfrac{7}{10}kt)$$

M1 A1ft

When $t = 0$, $\frac{dx}{dt} = 0$, $0 = \frac{k}{10}A + \frac{7}{10}kB$

M1

$$B = \frac{a}{14}$$

$x = \frac{a}{14} e^{-\frac{k}{10}t}$ (7 cos $\frac{7}{10}kt$ + sin $\frac{7}{10}kt$)

A1

9

[19]

Q15.

(a) CF $\ddot{x} + \dot{x} = 0$

$x = Ae^{nt}$ $n^2 + n = 0$

M1

$n = 0, -1$

$x = A + Be^{-}$

A1

PI $x = C \cos 2t + D \sin 2t$

Need both terms

M1

$-4C \cos 2t - 4D \sin 2t - 2C \sin 2t + 2D \cos 2t = k \sin 2t$

$-4C + 2D = 0$ and $-4D - 2C = k$

$C = -\frac{k}{10}$; $D = -\frac{k}{5}$

A1 if at least one correct

A1

$x = A + Be^{-t} - \frac{k}{10} \cos 2t - \frac{k}{5} \sin 2t$

ft dep on M2 above

A1ft

$t = 0, x = a$ $a = A + B - \frac{k}{10}$

ft dep on M2 above

A1ft

$\dot{x} = -Be^{-t} + \frac{k}{5} \sin 2t - \frac{2k}{5} \cos 2t$

M1ft

$t = 0, \dot{x} = 0;$ $0 = -B - \frac{2k}{5}$ $B = -\frac{2k}{5}$

A1

$$A = a + \frac{k}{10} - B$$

$$A = a + \frac{k}{2}$$

A1

$$x = a + \frac{k}{2} - \frac{2k}{5}e^{-t} - \frac{k}{10}\cos 2t - \frac{k}{5}\sin 2t$$

A1

11

(b) If e^{-t} term may be ignored, range of $\frac{k}{10}\cos 2t + \frac{2k}{10}\sin 2t$ using $a\cos\theta + b\sin\theta = R\cos(\theta - \alpha)$

M1

is $\pm\frac{k\sqrt{5}}{10}$

Max / min

M1 A1

$\therefore$ values are

ft dep on all M gained in (a)

A1ft

4

$a + \frac{k}{2} - \frac{k\sqrt{5}}{10}$ and $a + \frac{k}{2} - \frac{k\sqrt{5}}{10}$

if differentiation used, ***sc*** *2 for either*

These are $a + 0.276k$ and $a + 0.7236k$

[15]

Q16.

(a) $mg = \frac{\lambda e}{l}$

Equation for equilibrium

M1

$e = \frac{mgl}{\lambda}$

Correct extension

A1

2

(b) $T = -\frac{\lambda}{1}\left(x + \frac{mgl}{\lambda}\right)$

Expression for tension

Correct expression

A1

$m\frac{d^2x}{dt^2} = mg - T$

Application of Newton's 2nd law with T and mg

m1

$= mg - \frac{\lambda}{1}\left(x + \frac{mgl}{\lambda}\right)$

Simplified to correct final answer

A1

5

(c) $\omega = 2\pi\sqrt{\frac{\lambda}{ml}}$

$P = 2\pi\sqrt{\frac{ml}{\lambda}}$

Finding period

M1

Correct final answer

A1

2

(d) Increased by a factor of 2

Increased

B1

Factor of 2

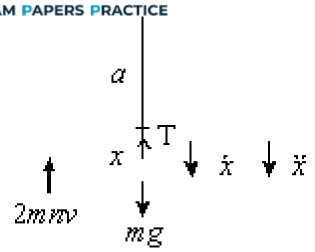
B1

2

[11]

Q17.

(a)



Using $F = ma$,

$$m\ddot{x} = mg - T - 2mnv$$

M1

$$= mg - 6mn^2 a, \frac{x}{a} - 2mnv$$

A1

$$m\ddot{x} + 6mn^2 x + 2mnx = mg$$

$$i.e. \quad \frac{d^2x}{dt^2} + 2n\frac{dx}{dt} + 6n^2 x = g$$

A1

3

(b) Substituting $x = Ae^{pt}$

M1

$$p^2 + 2np + 6n^2 = 0$$

$$p = \frac{\left(-2n \pm \sqrt{4n^2 - 24n^2}\right)}{2}$$

$$= -n \pm \sqrt{5}\, i\, n$$

A1

$\therefore$ CF is $x = e^{-nt}(A\cos\sqrt{5}\,nt + B\sin\sqrt{5}\,nt)$

A1

PI is $x = \frac{g}{6n^2}$

B1

$\therefore$ GS is $x = e^{-nt}$ (A cos $\sqrt{5}\,nt + B\sin\sqrt{5}\,nt$

$= \frac{g}{6n^2}$

A1

When $t = 0$, $x = 0 \Rightarrow 0 = A + \frac{g}{6n^2}$

M1

$\therefore A = -\frac{g}{6n^2}$

A1

When $t = 0$, $\frac{dx}{dt} = 0$;

$\frac{dx}{dt} = -ne^{-nt}(A\cos\sqrt{5}nt \times B\sin\sqrt{5}nt)$

$+ e^{-nt}\left(-\sqrt{5}n\,A\sin\sqrt{5}nt + \sqrt{5}n\,B\cos\sqrt{5}nt\right)$

$+e^{-nt}\left(-\sqrt{5}n\ \ A\sin\sqrt{5}nt + \sqrt{5}n\ \ B\ \cos\sqrt{5}nt\right)$

M1

$\Rightarrow 0 = -n\,A \times \sqrt{5}nB)$

$\therefore B = \frac{1}{\sqrt{5}}A = -\frac{g}{6\sqrt{5}n^2}$

A1

$\therefore x = \frac{g}{6n^2} -$

$\left(\frac{g}{6n^2}\cos\sqrt{5}nt + \frac{g}{6\sqrt{5}n^2}\sin\sqrt{5}nt\right)e^{-nt}$

A1

10

(c) $\frac{dx}{dt} =$

$ne^{-nt}\left(\frac{g}{6n^2}\cos\sqrt{5}nt + \frac{g}{6\sqrt{5}n^2}\sin\sqrt{5}nt\right) -$

$$e^{-nt}\left(-\frac{\sqrt{5}g}{6n}\sin\sqrt{5}nt+\frac{g}{6n}\cos\sqrt{5}nt\right)$$

M1

$$\frac{dx}{dt}=0 \text{ when } \cos\sqrt{5}nt+\frac{1}{\sqrt{5}}\sin\sqrt{5}nt+$$

$$\sqrt{5}\sin\sqrt{5}nt\cos\sqrt{5}nt=0$$

$$\sin\sqrt{5}nt=0$$

M1A1

First time at rest is when $t = \frac{\pi}{\sqrt{5}n}$

When $t = \frac{\pi}{\sqrt{5}n}$, $\frac{dx}{dt}=0$ SC2

A1

4

[17]

Q18.

(a) $t = \frac{1}{4} \times \frac{2\pi}{5}$

$\omega = 5$

$\frac{1}{4}$ *of their period*

B1 M1

$= \frac{\pi}{10}$

A1

3

(b) $v_{max} = 5a$

$v_{max} = 5a$

B1

$$\left(\frac{5a}{2}\right)^2 = 5^2\left(a^2-x^2\right)$$

Use of $v^2 = \omega^2(a^2 - x^2)$ *with* $v = \frac{5a}{2}$

M1

$$x^2 = \frac{3a^2}{4}$$

Correct equation

Solving for x

A1 m1

$$x = \frac{a\sqrt{3}}{2}$$

Answer as given or equivalent

A1

5

[8]

Q19.

(a) Substituting $z = Ae^{nt}$,

$2n^2 + 6n + 9 = 0$

M1

$$n = \left(-6 \pm \frac{\sqrt{36-72}}{4}\right)$$

$$n = -\frac{3}{2} \pm \frac{3}{2}\text{i}$$

A1

$$\therefore\ x = e^{-\frac{3}{2}t}\left(A\cos\frac{3}{2}t + B\sin\frac{3}{2}t\right)$$

M1 A1ft

When $t = 0$, $x = a$; $a = A$

B1

Differentiating

$$\frac{dx}{dt} = -\frac{3}{2}e^{-\frac{3}{2}t}\left(A\cos\frac{3}{2}t + B\sin\frac{3}{2}t\right) + \frac{3}{2}e^{-\frac{3}{2}t}\left(-A\cos\frac{3}{2}t + B\sin\frac{3}{2}t\right)$$

Use of product formula

M1 A1

When $t = 0$, $\frac{dx}{dt} = 0$; $0 = -\frac{3}{2}A + \frac{3}{2}B$

M1

$B = a$

$\therefore\ x = ae^{-\frac{3}{2}t}\left(\cos\frac{3}{2}t + \sin\frac{3}{2}t\right)$

A1

9

(b) Damping motion is light damping

And one reason

B1

since the particle oscillates, with its amplitude reducing to zero

B1

2

Or

e^{-kt} × trig. term

Or

Sketch from graphics calculator

Full reason

[11]

Q2.

This question was answered well in most cases. Part (a) was popular and mostly yielded full marks. Solutions to the auxiliary equation in part (b)(i) were often fully correct, with the most frequent error being the omission of 'i' in the solution, despite the obvious nature of the roots. Some chose a verification method for part (b)(ii) but only consideration of a general solution was sufficient for full credit. Solutions in part (c) were impressive.

Q3.

Again, this question was answered well, with candidates showing good knowledge and understanding of this topic. The appropriate techniques were usually applied efficiently to produce correct solutions.

Q4.

Part (a) was completed successfully by most candidates. Part (b) was often successfully answered, with a common error being the omission of the weight and an incorrect expression for the tension in the general position; often these two errors cancelled each other out in subsequent working, leading to an apparently correct solution. The main error in part (c) was to stop work having found the full general solution and not to evaluate the remaining constants. Part (d) was not answered well.

Q5.

Weaker candidates created the printed differential equation in part (a) by dubious means. It was necessary to find the equilibrium position and then use the extensions to be $2a + x$ in string AP and $a - x$ in string BP. Part (b) was answered very well by most candidates.

The only common error was in trying to insert the boundary conditions and finding $\frac{dx}{dt}$, when $\frac{d}{dx}\cos\sqrt{2}t$ sometimes became $-\sin\sqrt{2}t$ rather than $-\sqrt{2}\sin\sqrt{2}t$.

Q6.

Part (a) was answered well, although again some candidates were clearly working back from the printed result to obtain the initial equation of motion. Part (b) was attempted well while the common error was in using $x = a$ when $t = 0$, not noticing that x was the distance below A, and initially the particle was released at A; thus α = 0 when $t = 0$. Relatively few candidates were totally accurate in finding the velocity of the particle in part (c) and only these successful candidates could notice that the terms in cos $\sqrt{3}nt$ cancelled out, leaving only the terms containing sin $\sqrt{3}nt$.

Q7.

The great majority of candidates were able to use Hooke's law and Newton's second law to answer parts (a) and (b) of this question. Some candidates were unable to explain why $\dot{x} = 1 - v$ for part (c)(i) but were still able to gain full marks for part (c)(ii).

Some candidates were able to answer part (d) of the question without answering part (c).

Even though only the general solution for the differential equation was requested, some candidates proceeded to find the particular solution also. Many candidates were able to state the limiting value of x requested in part (e).

Q8.

A relatively large number of candidates had difficulty drawing the force/displacement diagram requested for part (a) of this question. A great majority of the candidates were able to answer part (b) correctly even when some of these candidates had not drawn a correct diagram.

The responses to part (c) were generally good, indicating a high level of knowledge and skills on the part of candidates in solving second order differential equations where the roots of the auxiliary equation are complex.

Part (d) of this question proved too difficult for some candidates.

Q9.

It was pleasing to see good responses from the great majority of candidates to this question. Candidates showed a high degree of competence and confidence in solving the second order differential equation and in interpreting their solutions to part (c).

Q10.

The candidates did well with part (a) and for quite a number of candidates these were the only marks that they obtained on this question.

In part (b), the majority of the candidates totally ignored the weight of the sphere and produced invalid arguments. There were some candidates who could produce the required result without difficulty.

In the last part, there were quite a few candidates who found the period, but did not realise that the time needed was a quarter of the period.

Q11.

This question was answered well. Apart from a few arithmetical errors most candidates obtained correctly $x = (18\cos t + 51\sin t)e^{-3t} + \sin 3t - 18\cos 3t$. Many candidates appreciated that, when t becomes large, the e^{-3t} term became negligible but only a few described fully, as required, the resulting expression for x, $\sin 3t - 18\cos 3t$.

Q12.

Part (a) of this question was done very well by almost all of the candidates. Part (b)(i) was very much more demanding and there were very few correct solutions. Most of the candidates did not find a two term expression for the tension in the string and did not include gravity when applying Newton's second law. In contrast, part (b)(ii) was done quite well, although some candidates did have difficulty simplifying their results.

Q13.

In part (a), a significant proportion of candidates worked back from the given result. Others used x as the position beyond the equilibrium position rather than as stated, and then had to 'lose' terms. In part (b), candidates made very good progress with many obtaining the correct result. The common error was forgetting an "n' in the differentiation of one of the terms when finding $\frac{\mathrm{d}x}{\mathrm{d}t}$ to insert a boundary condition. Other candidates could not simplify correctly the term $\frac{-2n \pm \sqrt{-4n^2}}{2}$.

Q14.

Most candidates answered part (a) correctly but a number did not use $mg + T_{PB} = T_{AP}$ and fiddled the algebra to give the printed result, for which they gained no marks.

In part (b), candidates had to find tensions in the two springs AP and PB, with extensions $(3a + x)$ and $(a - x)$ respectively, and then use $F = ma$ downwards, giving $m\ddot{x} = mg + T_{PB} - T_{AP} - \frac{1}{5}mkx$. Without these equations candidates were not given credit for producing the printed differential equation.

All candidates used the printed equation given in part (b)(i) to find x in part (ii). A number of candidates lost k in a variety of places, but virtually all obtained a result similar to that required.

Q15.

The complementary function $x = A + Be^{-t}$ caused difficulty to some candidates after they had obtained $n = 0$ and $n = -1$ from their initial substitution of $x = A\,e^{nt}$. A number then wrote $x = (A + Bt)\,e^{-t}$.

The particular integral $x = C\cos 2t + D\sin 2t$ was also often simplified to $x = C\sin 2t$, with candidates ignoring the cos $2t$ term that they had obtained in their differentiation. Those who used the correct form of the complementary function and the correct form of the particular integral usually obtained the correct general solution.

After making the e^{-t} term go to zero in part (b), some used differentiation to find the maximum and minimum values but this was very time consuming and offered many opportunities for making errors.

Better candidates appreciated that the range in values of $\frac{k}{10}\cos 2t + \frac{k}{5}\sin 2t$ was $\pm\sqrt{\left(\frac{k}{10}\right)^2 + \left(\frac{k}{5}\right)^2}$ from the use of $a\cos\theta + b\sin\theta = R\cos(\theta - \alpha)$.

Q16.

There were many good answers to part (a). Virtually all those who realised that they needed to consider when the forces were in equilibrium gained both marks. Part (b) was found to be very much more difficult. Several candidates did not include the weight of the particle at any stage of their solutions. Some others produced very contrived arguments to produce the negative sign in the differential equation. Part (c) was often done well, and those who did not gain marks on this question did not seem to know how to proceed. In part (d), several of the candidates stated that the period would increase, but a smaller number were able to quantify the increase correctly.

Q17.

Many candidates attempted to find the extension of the string in the equilibrium position, although in this question it was not relevant. These candidates had no mg force present when they found their differential equation. As this equation was printed on the question paper, candidates who then 'created' the mg force rarely scored marks in part (a). In part (b), candidates usually obtained the general solution $x = e^{-nt}(A\cos\sqrt{5}nt + B\sin\sqrt{5}nt)$, and most attempted to find the particular solution. They then showed the correct method for obtaining the actual solution. Those who correctly found the answer to part (b) often substituted $t = \dfrac{\pi}{\sqrt{5}n}$ to show that the particle was at rest.

Only the better candidates differentiated ther answer to show that the particle was at rest.

Q18.

There were some good solutions to this question, but many candidates found it difficult to make much progress. Part (a) in particular caused a lot of difficulty. Quite a number of candidates did not realise that a quarter of the period was needed, often using a half instead. In part (b), there were more correct solutions, but some candidates did not seem to know how to approach the problem.

Q19.

Part (a) was completed well by most candidates. However the use of the formula to solve the quadratic equation $2n^2 + 6n + 9 = 0$ was completed incorrectly by a number of candidates. Most appreciated that the damping was light, although the reason given for this was often imprecise.